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Serial No: RA-20-0261
August 27, 2020

U.S. Nuclear Regulatory Commission
Washington, D.C. 20555
ATTENTION: Document Control Desk

10 CFR 50.73

Subject: Duke Energy Carolinas, LLC
McGuire Nuclear Station, Unit 1
Docket No. 50-369, 50-370
Renewed License No. NPF-9, NPF-17
Licensee Event Report 2020-01, Revision 0
Nuclear Condition Report Number 02337223

Pursuant to 10 CFR 50.73 Sections (a)(1) and (d), attached is Licensee Event Report (LER) 2020-01, Revision 0, regarding Condition Prohibited by Technical Specifications.

This report is being submitted for Unit 1 and 2 in accordance with 10 CFR 50.73(a)(2)(i)(B), "Operation or Condition Prohibited by Technical Specifications." The Technical Specification (TS) to which the requirements of section (a)(2)(i)(B) applies is TS 3.4.15, RCS Leakage Detection Instrumentation.

This event is considered to have no significance with respect to the health and safety of the public. There are no regulatory commitments contained in this LER.

If questions arise regarding this LER, contact Joseph F. Hussey at 980-875-5045.

Sincerely,

Thomas D. Ray
Site Vice President
McGuire Nuclear Station

Attachment

TEZZ
NRR

www.duke-energy.com

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**LICENSEE EVENT REPORT (LER)**

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R 3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-8 A10M), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to infocoll@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503, e-mail: omb-submissions@omb.eop.gov. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. Facility Name

McGuire Nuclear Station Unit 1

2. Docket Number

05000369

3. Page

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4. Title

Condition prohibited by Technical Specification 3.4.15, Reactor Coolant System (RCS) Leakage Detection Instrumentation

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Rev No.	Month	Day	Year	Facility Name	Docket Number
06	29	2020	2020	01	00	08	27	2020	McGuire Nuclear Station Unit 2	05000370
									Facility Name	Docket Number
										05000

9. Operating Mode	11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)			
1	☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)
	☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)
	☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)
	☐ 20.2203(a)(2)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
10. Power Level	☐ 20.2203(a)(2)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(v)(A)	☐ 73.71(a)(4)
100	☐ 20.2203(a)(2)(iii)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(5)
	☐ 20.2203(a)(2)(iv)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(1)
	☐ 20.2203(a)(2)(v)	☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(i)
	☐ 20.2203(a)(2)(vi)	☒ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(vi)	☐ 73.77(a)(2)(ii)
		☐ 50.73(a)(2)(i)(C)	☐ Other (Specify in Abstract below or in NRC Form 368A)	

12. Licensee Contact for this LER

Licensee Contact: Joseph Hussey

Telephone Number (Include Area Code)
(980) 875-5045**13. Complete One Line for each Component Failure Described in this Report**

Cause	System	Component	Manufacturer	Reportable to ICES	Cause	System	Component	Manufacturer	Reportable to ICES
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14. Supplemental Report Expected☐ Yes (If yes, complete 15. Expected Submission Date) ☒ No**15. Expected Submission Date****Abstract**

On June 29, 2020 with Unit 1 in Mode 1 at approximately 100 percent power and Unit 2 in Mode 1 at approximately 100 percent power, McGuire determined that Revision 0 of calculation MCC-1223.15-01-0015, Containment Floor and Equipment Sumps Volume Calculation, contained non-conservative assumptions that would have prevented the containment floor and equipment (CFAE) sump level monitors from detecting and alarming on a one gallon per minute input within one hour. Revision 0 of calculation MCC-1223.15-01-0015 was approved for use on April 15, 1994.

Discovery of the non-conservative assumptions in Revision 0 of calculation MCC-1223.15-01-0015 resulted in a condition prohibited by Technical Specification (TS) 3.4.15, Reactor Coolant System (RCS) Leakage Detection Instrumentation, since the CFAE Sump Monitors for both Units was determined to be inoperable for longer than allowed by the TS 3.4.15 Limiting Condition for Operation (LCO) action statements.

No causal analysis was performed due to the legacy nature of Revision 0 of calculation MCC-1223.15-01-0015.

An Engineering Change has been implemented to Insert the revised containment floor and equipment sump monitor rate of rise set point, from revised calculation MCC-1223.15-01-0015, on the Operator Aid Computer.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R 3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-6 A10M), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and the OMB reviewer at OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503, e-mail: omb_submission@omb.eop.gov. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
McGuire Nuclear Station	05000369	2020	01	00

NARRATIVE**BACKGROUND**

The following information is provided to assist readers in understanding the event described in this LER. Applicable Energy Industry Identification [EII] System and component codes are enclosed within brackets. McGuire Nuclear Station unique system and component identifiers are contained within parentheses.

Liquid Waste Recycle [WD] (WL), [LIT] (LT), Containment Floor and Equipment (CFAE) Sump Level Monitors

The CFAE Sump Level Monitors function by taking the CFAE sump level change, rate of rise, as measured by the level transmitters (1/2WLLT5250 and 5260) and calculated by the Operator Aid Computer to provide indication in the Control Room of possible leakage from Reactor Coolant System [AB] (NC), Main Steam System [SB] (SM), Main Feedwater System [SJ] (FW), or Auxiliary Feedwater System [BA] (CA). The sump level monitor is capable of detecting a 1 gpm leak in 1 hour. The sensitivity is based upon the ability of the sump level indication to sense a change in level of 1/4 inch and the following assumptions

- Sump level indication is between 5.75 to 17 inches
- Sump level indication moves in quarter-inch increments
- Operator Aid Computer rate alarm set at 0.0167 in/min

Operator Aid Computer [ID] (IKE)

The Operator Aid Computer System (OAC) at the McGuire Nuclear Station is a non-QA system used to monitor parameters and events in various plant systems and to provide data and alarms to the Control Room Operators and other station personnel. The data and alarms provided by the OAC are important to the Control Room Operators' ability to perform routine surveillances and other required activities during normal operation, as well as to respond quickly to changing plant conditions during transients and abnormal events. The OAC performs the CFAE Sump Level Monitor function as described below.

CFAE Sump Level Instrumentation is used to monitor changes in level due to inputs and/or pump downs of the sump as necessary. To convert the change in sump level values into units that can be directly correlated to leakage, OAC algorithms were developed. The algorithms are based on functional relationships (e.g. geometry and mathematics) and rate of change transform calculations. The algorithm uses standard computer point types and predefined transform and logic functions on the OAC to convert the level instrumentation signals for comparison against a calculated threshold. The output of the algorithm is a displayed rate value for both sumps that is updated every minute on the OAC. The algorithm includes rate alarm logic which is processed every 5 seconds. The OAC alarm threshold is selected to detect a total leakage rate change of 1 GPM within one hour.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

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McGuire Nuclear Station	05000369	2020	01	00

NARRATIVE**EVENT DESCRIPTION**

On June 29, 2020 with Unit 1 in Mode 1 at approximately 100 percent power and Unit 2 in Mode 1 at approximately 100 percent power, McGuire determined that Revision 0 of calculation MCC-1223.15-01-0015, Containment Floor and Equipment Sumps Volume Calculation, contained non-conservative assumptions that would have prevented the containment floor and equipment (CFAE) sump level monitors from detecting and alarming on a one gallon per minute input within one hour. Revision 0 of calculation MCC-1223.15-01-0015 was approved for use on April 15, 1994.

Discovery of the non-conservative assumptions in Revision 0 of calculation MCC-1223.15-01-0015 resulted in a condition prohibited by Technical Specification (TS) 3.4.15, Reactor Coolant System (RCS) Leakage Detection Instrumentation, since the CFAE Sump Monitors for both Units was determined to be inoperable for longer than allowed by the TS 3.4.15 Limiting Condition for Operation (LCO) action statements.

REPORTABILITY DETERMINATION

10 CFR 50.73(a)(2)(i)(B) requires, in part, the licensee to report "Any operation or condition which was prohibited by the plant's Technical Specifications...". As the CFAE sump level monitors rate of rise set point was derived from Revision 0 of calculation MCC-1223.15-01-0015 the containment floor and equipment sump level monitors have been inoperable since April of 1994.

CAUSAL FACTORS

No causal analysis was performed due to the legacy nature of Revision 0 of calculation MCC-1223.15-01-0015.

CORRECTIVE ACTIONS

Immediate Action:

1. Operations declared the CFAE sump monitor on both Units Inoperable.

Interim Action:

1. Engineering revised calculation MCC-1223.15-01-0015. (Complete)
2. McGuire Implemented temporary OAC alarm on each Unit using the new containment floor and equipment sump monitor rate of rise set point from revised calculation MCC-1223.15-01-0015 to return the monitors to OPERABLE. (Complete)

Subsequent Action:

1. An Engineering Change has been implemented to insert the revised containment floor and equipment sump monitor rate of rise set point, from revised calculation MCC-1223.15-01-0015, on the Operator Aid Computer. (Complete)

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

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NARRATIVE**SAFETY ANALYSIS****Safety Analysis**

The safety impact associated with this issue is negligible for the following reasons:

- Alternate means of detecting small leaks were available using the Containment Atmosphere Particulate Radioactivity monitor, Ventilation Unit Condensate Drain Tank level indication, Incore Instrument Sump level indication, along with frequent containment sump level monitoring and RCS leak rate calculations. Because leakage into the containment sump is typically very low, an abnormal RCS leakage of one gallon per minute would be quickly recognized. If the leak rate was more than 1.5 gallons per minute, the CFAE sump level monitors would have gone into alarm and performed their function to alert Control Room operators.
- RCS leaks of this non-alarming range between 1.0 and 1.5 gpm are not risk significant in the McGuire PRA, as small RCS leaks are well within the capacity of normal make-up pump flowrates and are not expected to challenge the capability of plant systems to achieve an otherwise normal and safe plant shutdown.

As a result, the event is of minimal safety significance since the alarm setpoint was only slightly non-conservative and other diverse means of detection were available. Therefore, it is concluded that the impact of this issue had no impact on public health and safety.

ADDITIONAL INFORMATION

A search of the Corrective Action Program (NCR) database was conducted, based on combinations of keywords, to determine if this event was recurring at McGuire, i.e., similar significant event with the same keywords. Also No NCR of similar significance with the same keywords was identified. As such, this is not considered a recurring event.